EXP.NO:13

FINDING FLOOR VALUE

PROBLEM STATEMENT:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

CODE:

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int x;
    scanf("%d",&x);
    int num=arr[0];
    for(int i=1;i<n;i++){
        if(arr[i]<=x && arr[i]>num){
            num=arr[i];
        }
    }
    if(num<=x){
        printf("%d",num);
    }
}
```

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OUTPUT:



	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.